An exact specified-state collision threshold in a binary automaton family

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9 November 2026

Abstract

For a specified state of a synchronizing automaton, the compress-with-another threshold is the shortest length of a word that merges it with any other state. We determine this threshold for the corrected binary family in Figure 3 of Szykuła's survey of open problems. For every $p \geq 3$, the family has $n = 3p$ states and the distinguished state satisfies $\mu(q_0) = 4p = 4n/3$, attained by $(ba)^p(ab)^p$. A self-contained pair-automaton proof establishes the lower bound using two entrance potentials and an excluded region, including the endpoint case $p = 3$. An explicit reset word of length $3p^2 + 4p - 1$ proves synchronization for every parameter. The family therefore realizes the proposed $4n/3$ specified-state threshold and requires a leading constant of at least $4/3$ in any uniform linear bound, even for binary strongly connected automata whose letters have corank one.

1 Introduction

A complete deterministic semiautomaton (Q, Σ, δ) is synchronizing if some word maps Q to a singleton. Such a word is a reset word, and the reset threshold $\mathrm{rt}(\mathcal{A})$ is its minimum length. We write transitions as right actions, $qw = \delta(q, w)$. For a state q, put

$$\mu(q) = \min\{|w| : qw = rw \text{ for some } r \in Q \setminus \{q\}\}.$$

The minimum is finite in a synchronizing automaton. It concerns one collision involving a specified initial state, whereas a reset word must merge all initial states.

Szykuła [4] asks whether $\mu(q) = O(n)$ in every strongly connected synchronizing n-state automaton. His Figure 3, attributed there to Dżyga's 2018 master's thesis [3], gives a binary family on $3k + 6$ states proposed to attain $4n/3$. Conjecture 8 asserts synchronization of the family and this specified-state threshold. For general background we refer to Volkov [2]; the Černý conjecture [1], asking for reset words of length at most $(n-1)^2$, remains open. We prove the proposed threshold and synchronization for the completed transition definition below.

We use the equivalent parameter $p = k + 2 \geq 3$. The figure in [4] leaves some boundary transitions implicit; Section 2 supplies a complete transition definition. The word printed in Conjecture 8, $(ab)^p a(ab)^p$, has length $4p + 1$. In contrast, the word $(ba)^p(ab)^p$ merges q_0 with q_2 in exactly $4p$ letters. Our interpretation of the original question follows the EPTCS publication and the public arXiv v1 of [4].

We track a pair of distinct states until its first collision. Each letter merges only one pair, so the threshold reduces to the distance from an initial pair containing q_0 to one of these two kernel pairs. Two time lower bounds, propagated across complete lists of incoming transitions, show that neither kernel pair can be reached earlier than claimed. An invariant excludes otherwise possible shortcuts. The argument applies to arbitrary words and initial partners; a separate image-shrinking construction proves synchronization.

An earlier release by Korea Superintelligence Labs, generated by Machina Mathematica [5], established the all-parameter upper witness and reported exact finite computations. Lawson subsequently announced and published a candidate all-parameter proof, including a pair-space certificate and a separate synchronization construction [6, 7]. His displayed reset word has length $4p^2 + 6p$; the construction here gives the smaller bound $3p^2 + 4p - 1$. Szykuła has also informed Qichao Wang of other independently communicated proofs. Alongside these developments, this article gives a self-contained proof based on complete entrance classifications and two potential bounds, together with the explicit shorter reset construction.

2 The automaton family $\mathcal{A}_p$

Let $\mathcal{A}_p$ have $Q = \{q_0, \dots, q_{3p-1}\}$ and alphabet $\{a, b\}$. The complete definition is conveniently given by matchings and three exceptional edges. Letter a fixes q_0, swaps q_{2j-1} and q_{2j} for $1 \le j \le p-1$, and has

$$q_{2p-1}a = q_{2p}, \qquad q_{2p}a = q_{2p+1}, \qquad q_{2p+1}a = q_3.$$

On $q_{2p+2}, \dots, q_{3p-1}$ it swaps successive pairs starting at the even index $2p+2$; an unmatched last state is fixed. Letter b swaps q_{2j} and q_{2j+1} for $0 \le j \le p-1$, has $q_{2p}b = q_2$, and swaps successive pairs on $q_{2p+1}, \dots, q_{3p-1}$ starting at $2p+1$, again fixing an unmatched last state. In particular, $q_0b = q_1$ and $q_1b = q_0$. The initial b-transposition and the terminal matching edges make explicit the boundary transitions left implicit in the schematic of [4]. Throughout, word length is counted in the original alphabet $\{a, b\}$; the two-letter blocks introduced later are not additional alphabet letters.

For the proof write

$$M_i = q_i \ (0 \le i \le 2p), \qquad O_h = q_{4p-h} \ (p+1 \le h \le 2p-1).$$

The outer coordinate runs in the reverse direction. Away from the exceptions, on either branch letter a increases an odd coordinate by one and decreases an even coordinate by one; b does the opposite. At the outer bottom $h = p+1$, the decreasing letter fixes the state. It is a when $p+1$ is even and b when $p+1$ is odd. The remaining exceptional rules are

$$M_0a = M_0, \quad M_{2p}a = O_{2p-1}, \quad M_{2p}b = M_2, \quad O_{2p-1}a = M_3. \tag{1}$$

Both letters have rank $3p-1$. Indeed, their only double fibers are

$$a^{-1}(M_3) = \{M_4, O_{2p-1}\} = K_a, \qquad b^{-1}(M_2) = \{M_3, M_{2p}\} = K_b. \tag{2}$$

Each letter misses one image state: a misses M_{2p-1} and b misses M_{2p}. All other fibers are singletons. The underlying directed graph is strongly connected. The main matchings connect $M_0, \dots, M_{2p-1}$ in both directions, the final main a-edge reaches M_{2p}, and a then enters the outer branch. Its matchings connect every outer state in both directions, and its top returns to M_3; $M_{2p}b = M_2$ also returns to the main chain.

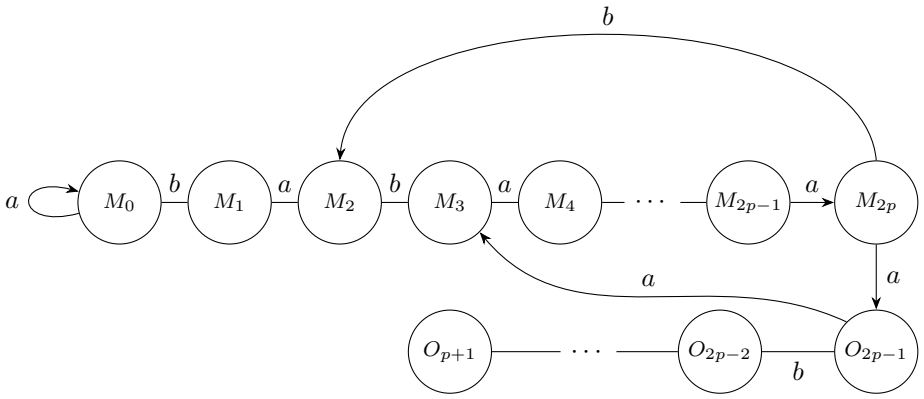


Figure 1: The transition structure of $\mathcal{A}_p$ (adapted from [4]), shown schematically for $p \geq 5$. A labelled line without arrowheads represents a two-way transposition; a directed arrow represents a one-way transition. Unlabelled segments continue the alternating matchings. The outer bottom O_{p+1} is fixed by its decreasing letter. For $p = 4$, the outer branch is O_7, O_6, O_5 with no intermediate vertex; for $p = 3$, it is O_5, O_4 and the displayed labels O_{2p-2} and O_{p+1} denote the same vertex. The transition rules in the text apply in every case.

3 Exact pair distances and the threshold

The unordered pair graph has vertex set $\binom{Q}{2}$. It has a directed edge $P \xrightarrow{x} Px$ for $x \in \{a, b\}$ whenever $|Px| = 2$; collapsing edges are omitted. Set

$$S_p = \{\{M_0, q\} : q \in Q \setminus \{M_0\}\}.$$

Distance from S_p means the minimum over its vertices. A first collision must use a kernel pair, so

$$\mu(q_0) = 1 + \min\{\mathrm{dist}(S_p, K_a), \mathrm{dist}(S_p, K_b)\}. \tag{3}$$

Lemma 1. *For every $p \geq 3$,*

$$\mathrm{dist}(S_p, K_b) = 4p - 1, \qquad \mathrm{dist}(S_p, K_a) = 4p.$$

Proof. We prove the lower bounds first. Fix an arbitrary path $P_0 \xrightarrow{x_1} P_1 \xrightarrow{x_2} \cdots \xrightarrow{x_\ell} P_\ell$ in the distinct-pair graph, where $P_0 \in S_p$ and $x_t \in \{a, b\}$. All time bounds refer to its vertex P_t after t original letters.

1. The index bound and excluded region. Every original state index increases by at most one in one letter. Thus the trajectory starting at q_0 can have index no greater than the elapsed time t. Since one member of P_t is this trajectory, we obtain

$$t \geq \min\{i : q_i \in P_t\}. \tag{4}$$

In particular, for a mixed pair $P_t = \{M_x, O_h\}$ the original indices are x and $4p - h > 2p \geq x$, respectively. Thus (4) implies $t \geq x$ directly, without converting the outer coordinate back at every occurrence.

Let J consist of opposite-parity main pairs whose coordinate sum exceeds $2p + 1$. The only reachable member of J is K_b. To see this, an ordinary matching step on an opposite-parity pair

preserves its sum, and an ordinary step on a same-parity pair preserves that parity type. A pair involving the fixed M_0 cannot acquire a sum above $2p+1$. The b-jump from M_{2p} lands at M_2; its other main image is at most M_{2p-1}. Hence the resulting main-pair coordinate sum is at most $2+(2p-1)=2p+1$, so it cannot enter J, which requires a strictly larger sum. The a-step from M_{2p} leaves the main branch.

The terminal edge $M_{2p-1}a = M_{2p}$ also requires attention. A target in J containing M_{2p} has the form $\{M_{2p}, M_j\}$ with odd $j \geq 3$. For $5 \leq j \leq 2p-3$, its only a-preimage is $\{M_{2p-1}, M_{j+1}\}$, already in $J \setminus \{K_b\}$ since its sum is $2p+j > 2p+1$. The remaining odd endpoint $j = 2p-1$ has no a-preimage. For $j = 3$, the preimage using M_4 is likewise already in $J \setminus \{K_b\}$, while $\{M_{2p-1}, O_{2p-1}\}a = K_b$. An outer-to-main step lands at M_3, and an opposite-parity sum above $2p+1$ requires its other image to be M_{2p}, exactly the case just treated. Two outer states cannot both enter the main branch in one letter. Finally, $K_b a = K_a$, whereas $|K_b b| = 1$. No initial pair lies in J, and no edge from outside $J \setminus \{K_b\}$ enters that set. This proves the invariant by induction. Call a pair *admissible* if it is outside $J \setminus \{K_b\}$.

2. The first entrance potential. Put

$$C = \{\{M_x, M_{2p+1-x}\} : 1 \leq x \leq p\}, \qquad \phi_C(\{M_x, M_{2p+1-x}\}) = 2p - 2 + |x-3|.$$

No initial pair lies in C. Each internal edge changes the smaller coordinate by at most one, so the potential increases by at most one. The complete list of admissible entrances from outside C is

Letter	Source	Target	Source-time bound
a	$\{M_{2p-3}, O_{2p-1}\}$	$\{M_3, M_{2p-2}\}$	$2p-3$
b	$\{M_{2p-2}, M_{2p}\}$	$\{M_2, M_{2p-1}\}$	$2p-2$

Here and below source-time bounds concern arrival before the entering letter. Both follow from (4); after adding one they equal the target potential. For completeness of the list, an ordinary inverse matching preserves the opposite-parity sum, hence gives a source in C. The two exceptional inverse fibers account for the displayed sources: the a-fiber at M_3 yields the first row, and the b-fiber at M_2 yields the second. No additional entrance arises, since M_{2p} has no b-preimage, M_{2p-1} has no a-preimage, and the fixed state M_0 does not belong to C. This exhausts the exceptional cases. Thus induction on the path, including any exits and reentries, proves

$$P_t \in C \implies t \geq \phi_C(P_t), \qquad \operatorname{dist}(S_p, \{M_1, M_{2p}\}) \geq 2p. \tag{5}$$

3. The reflected entrance potential. Consider the disjoint tracks

$$A = \{\{M_x, O_{2p+1-x}\} : 2 \leq x \leq p\}, \qquad E = \{\{M_x, O_x\} : p+1 \leq x \leq 2p-1\}.$$

For $H = A \cup E$ set $\phi_H = 2p - 1 + x$, where x is the main coordinate. No initial pair lies in H. Ordinary internal edges change x by one. At the outer bottom, its decreasing letter exchanges A at $x = p$ with E at $x = p+1$; this also changes the potential by one. All admissible entrances from outside H are

Letter	Parameter	Source	Target	Source-time bound
a	all p	$\{M_1, M_{2p}\}$	$\{M_2, O_{2p-1}\}$	$2p$
b	all p	$\{M_{2p}, O_{2p-2}\}$	$\{M_2, O_{2p-1}\}$	$2p$
a	$p \geq 4$	$\{O_{2p-1}, O_{2p-3}\}$	$\{M_3, O_{2p-2}\}$	$2p+1$
a	$p = 3$	$\{O_5, O_4\}$	$\{M_3, O_4\}$	7

The first bound is (5); the others follow from (4). Adding one gives ϕ_H at each target.

We check completeness uniformly. In the interior of A, the coordinates have opposite parity, so both forward and inverse matchings preserve their sum. In the interior of E, the equal coordinates move together. At the outer bottom the fixed coordinate joins precisely the two tracks just described. At the outer top, $a^{-1}(O_{2p-1}) = \{M_{2p}\}$: target main coordinate 2 gives the first entrance, whereas the top target in E has main coordinate $2p-1$ and no a-preimage. Under b, the double fiber at M_2 gives the second entrance; its other source stays in H. To enter from two outer states under a, the main image must be M_3, hence the target is the $x = 3$ member of A. Its ordinary outer preimage is O_{2p-3} for $p \geq 4$. For $p = 3$, $h = 4$ is the bottom and is fixed by a, so that preimage is instead O_4. These are the last two rows. The missing main images and the remaining ordinary inverse matchings give no further entrances. Therefore

$$P_t = \{M_x, O_h\} \in H \implies t \geq \phi_H(P_t) = 2p - 1 + x, \qquad \operatorname{dist}(S_p, \{M_{2p-1}, O_{2p-1}\}) \geq 4p - 2. \tag{6}$$

For $p = 3$, the exceptional entrance reaches the member of A with $x = 3$, whose potential is $2p-1+3 = 8$. Likewise, the top of E has $x = 2p-1 = 5$, giving the lower bound $2p-1+5 = 10$.

4. Entrance into the kernel regions. There is only one admissible incoming edge to K_b:

$$\{M_{2p-1}, O_{2p-1}\} \xrightarrow{a} K_b. \tag{7}$$

Indeed, b misses M_{2p}, and the other a-preimage is $\{M_4, M_{2p-1}\}$, in $J \setminus \{K_b\}$. Equation (6) now yields $\operatorname{dist}(S_p, K_b) \geq 4p - 1$.

To control K_a define

$$B = \{\{M_x, O_{2p+3-x}\} : 4 \leq x \leq p+2\}, \qquad D = \{\{M_{h+2}, O_h\} : p+1 \leq h \leq 2p-2\}.$$

These tracks are disjoint. Their union U contains K_a and no initial pair. Its only admissible entrance from outside U is the a-transition from K_b to K_a. In fact, ordinary inverse matchings preserve the sum $2p+3$ in B, or the difference 2 in D. At the outer bottom the decreasing letter joins B at main coordinate $p+2$ to D at main coordinate $p+3$. The only U-target containing the outer top is the $x = 4$ member of B, namely K_a, whose a-preimage is K_b. A b-jump lands at M_2, and an outer-to-main a-jump at M_3, whereas every main coordinate in U is at least 4. Targets with main coordinate $2p$ have no b-preimage; those with main coordinate $2p-1$ have no a-preimage. The other top endpoint of D, namely $\{M_{2p}, O_{2p-2}\}$, has its ordinary a-preimage in D for $p \geq 4$, and in B for $p = 3$. Thus the list exhausts both endpoints and all exceptional fibers, also when $p = 3$ and D has only one member.

Every path reaching U has a first entrance immediately preceded by a visit to K_b. Hence its first entrance has length at least $4p$, and later visits cannot occur earlier. This proves $\operatorname{dist}(S_p, K_a) \geq 4p$. A final b-collapse costs at least $4p$, and a final a-collapse at least $4p+1$.

5. Matching paths and the upper word. Put $c = ba$ and $u = ab$. Direct substitution in the transition rules gives cycles, in the displayed order,

$$\begin{aligned} R &= (M_0, M_2, M_4, \ldots, M_{2p}, M_1) \quad \text{under } c, \\ T &= (M_0, M_1, M_3, \ldots, M_{2p-1}, M_2) \quad \text{under } u. \end{aligned} \tag{8}$$

Both have length $p+2$. In particular $\{M_0, M_2\}c^p = \{M_{2p}, M_1\}$ and $M_1 u^p = M_2$. The u-chain starting at M_{2p} visits the even-indexed outer states upwards, reflects at the last original state,

visits the odd-indexed outer states downwards, and then reaches M_2. It consists of M_{2p} and all $p-1$ outer states, so its entrance length is p. Consequently

$$\{q_0, q_2\}(ba)^p(ab)^p = \{q_2\}. \tag{9}$$

Immediately before the last a in this word, the pair is $\{M_{2p-1}, O_{2p-1}\}$. After that a it is K_b, and the final b collapses it. These pairs are distinct until the last letter: an earlier collision would persist and contradict the distinct penultimate pair. Thus the prefix of length $4p-1$ reaches K_b in the pair graph. Appending a, instead of the collapsing b, reaches K_a at length $4p$. This proves both upper distance bounds. □

Theorem 2. *For every $p \geq 3$, the automaton $\mathcal{A}_p$ defined in Section 2, with $n = 3p$ states, satisfies*

$$\mu(q_0) = 4p = \frac{4n}{3}.$$

The word $(ba)^p(ab)^p$ is a shortest word compressing q_0 with another state; it merges q_0 and q_2 at q_2.

Proof. Combine Lemma 1 with (3) and (9). □

This proves the specified-state threshold assertion of Conjecture 8 for the corrected family.

4 An explicit synchronization construction

A collision involving one state does not by itself establish synchronization. We now construct a reset word in three stages. First, a power of $c = ba$ maps the entire state set onto a $(p+2)$-cycle. Second, a preparation word produces a structured subset of this cycle, consisting of an interval of positions together with two endpoints. Finally, successive return words shorten this interval one position at a time, after which u^p, where $u = ab$, merges the last two states. This also explains the choice of the factors in the construction below. Identify the states of the c-cycle R in (8) with integer positions via the bijection

$$\rho : \{0, \ldots, p+1\} \longrightarrow R, \qquad \rho(0) = M_0, \quad \rho(i) = M_{2i}\ (1 \leq i \leq p), \quad \rho(p+1) = M_1.$$

For a set I of positions, write $\rho(I) = \{\rho(i) : i \in I\}$. Thus c adds one modulo $r = p+2$ to positions: $\rho(i)c = \rho((i+1) \bmod r)$.

The remaining main odd states descend by two under c until reaching M_1. The outer c-chain visits odd original indices upwards, reflects at the terminal state, visits even original indices downwards, and then reaches M_3 and M_1. Its largest distance into R is p. Since c permutes R, this proves the exact image identity

$$Qc^p = R. \tag{10}$$

For $j \geq p$, the return word $d_j = ac^j$ maps R into itself. Define its induced map on positions by $f_j(i) = \rho^{-1}(\rho(i)d_j)$. For $0 \leq i \leq p+1$, we have

$$f_j(i) = (j-i) \bmod r \quad (0 \leq i \leq p-1), \qquad f_j(p) = f_j(p+1) = (j+1) \bmod r. \tag{11}$$

For $1 \leq i \leq p-1$, $M_{2i}a = M_{2i-1}$ reaches M_1 after $i-1$ applications of c, giving the first formula. Position 0 is fixed by a. From position p, a reaches the outer top, which enters R at M_1 after p

applications of c. Position $p+1$ goes under a to M_2, giving the same residue. This verifies every position in (11).

For the preparation stage take

$$v_p = c^p u^{p-1} a^4 c^{p-2}, \qquad |v_p| = 6p-2.$$

The image of R under u^{p-1} is

$$\{M_0, M_1, M_3, \ldots, M_{2p-1}, O_{2p-1}\}.$$

This follows from the u-cycle and the u-chain used above. Under a^4, the two states M_{2p-1} and O_{2p-1} both reach M_4, while all remaining displayed states are fixed. The resulting image is $\{M_0, M_1, M_3, \ldots, M_{2p-3}, M_4\}$. Applying c^{p-2} gives, in R-positions,

$$Qv_p = \rho(I_{p-2}), \qquad I_t = \{0, \ldots, t\} \cup \{p, p+1\} \ (0 \le t \le p-2), \quad I_{-1} = \{p, p+1\}. \tag{12}$$

Indeed the odd states give positions $p-4, p-5, \ldots, 0, p+1$ when $p \ge 4$, while M_0, M_1, M_4 give $p-2, p-3, p$. For $p=3$, the prepared set before the final c is $\{M_0, M_1, M_3, M_4\}$; its image has positions $\{0,1,3,4\} = I_1$. Thus the same identity covers the smallest parameter.

For $0 \le t \le p-2$, equation (11) gives

$$f_{p+t}(I_t) = I_{t-1}. \tag{13}$$

For $2 \le t \le p-2$, the interval $\{0, \ldots, t\}$ maps to $\{(p+t-i) \bmod r : 0 \le i \le t\} = \{p, p+1, 0, \ldots, t-2\}$, where $r = p+2$. The pair $p, p+1$ merges to residue $(p+t+1) \bmod r = t-1$. Adding this image gives $\{p, p+1\} \cup \{0, \ldots, t-1\} = I_{t-1}$. At $t=1$, the interval maps to $\{p, p+1\}$ and the merged image is 0, giving I_0. At $t=0$, the interval maps to $\{p\}$ and the merged image is $-1 \bmod r = p+1$, giving $I_{-1} = \{p, p+1\}$. This checks the terminal steps as well as the general one.

Proposition 3. *For every $p \ge 3$, an explicit reset word for $\mathcal{A}_p$ is*

$$w_p = c^p u^{p-1} a^4 c^{p-2} (ac^{2p-2})(ac^{2p-3}) \cdots (ac^p) u^p. \tag{14}$$

It maps Q to $\{q_2\}$ and has length $3p^2 + 4p - 1$. In particular, $\mathcal{A}_p$ is synchronizing and

$$\mathrm{rt}(\mathcal{A}_p) \le 3p^2 + 4p - 1.$$

Proof. Equation (12) initializes the image. Applying (13) in descending order $t = p-2, \ldots, 0$ leaves positions $p, p+1$, that is $\{M_{2p}, M_1\}$. The final u^p maps both to M_2. Counting every a, b letter gives

$$|w_p| = (6p-2) + \sum_{j=p}^{2p-2} (2j+1) + 2p = (6p-2) + (3p-1)(p-1) + 2p = 3p^2 + 4p - 1. \qquad \square$$

Theorem 2 and Proposition 3 establish both assertions for the corrected family.

Corollary 4. *Suppose there are $\alpha \in \mathbb{R}$, $n_0 \in \mathbb{N}_{\ge 1}$ and a single function $\varepsilon : \mathbb{N}_{\ge 1} \to \mathbb{R}$ with $\varepsilon(n)/n \to 0$ such that, for every $n \ge n_0$, every binary strongly connected synchronizing n-state automaton whose letters have corank one, and every state q, one has $\mu(q) \le \alpha n + \varepsilon(n)$. Then $\alpha \ge 4/3$.*

Proof. The automata $\mathcal{A}_p$ belong to this class. For all sufficiently large p, the assumed uniform bound gives $4p \le 3p\alpha + \varepsilon(3p)$, hence $4/3 \le \alpha + \varepsilon(3p)/(3p)$. Letting $p \to \infty$ proves the claim. $\square$

5 Computational checks and reproducibility

The source code and reproduction commands are available in the public Figure 3 code archive (the `papers/figure3` and `scripts/direct_attack` directories). The programs independently reconstruct the transition tables and check the excluded region, complete lists of entrances, potential inequalities, collision witness, and reset-image identities at the 67 values $p = 3, \dots, 64$ and $100, 128, 256, 512, 1024$, reaching $n = 3072$ states. The test suite also checks that deliberately altered transition tables are rejected. Forward- and reverse-pair distance computations cover $p = 3, \dots, 12$; for $p = 3$, every nonempty binary word of length less than 12 is tested.

A second checker implements the transitions independently as piecewise functions and computes the exact distances to both kernel pairs for $p = 3, \dots, 128$, covering 1,812,783 reachable distinct pairs. It also checks the proposed shortest word, boundary cases, induced maps on cycle positions, preparation step, and each interval reduction. For $p = 3, \dots, 24$ the reset word is applied letter by letter to every state; for larger parameters it is checked by composing the individual transition maps. These finite checks corroborate the identities used in the argument; the all-parameter conclusions rest on the proofs above.

A related abstract entrance-bound lemma has been formalized in Lean 4.19.0 under explicit graph and entrance hypotheses. The concrete transitions of $\mathcal{A}_p$ and their all-parameter entrance classification have not been instantiated in Lean; thus the formalization is not a machine-checked proof of Theorem 2.

Acknowledgements. Qichao Wang thanks Marek Szykuła for correspondence.

AI Disclosure. OpenAI language-model tools, including Codex, assisted exploratory proof search, mathematical critique, and the computational checks underlying Sections 3–5. They also assisted the revision of the title, exposition, notation, diagram, and bibliography. This assistance included substantive review of the entrance-potential argument and reset construction, not only language editing. The all-parameter claims are supported by the explicit proofs; finite checks and the conditional Lean formalization have the limited roles stated above. Responsibility for the manuscript, including verification and approval of AI-assisted content, rests with the human authors.

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